

# Big Data Analytics Applications

Module 6

# What is PageRank?

- **Definition:**

- **PageRank** is an algorithm originally developed by Larry Page and Sergey Brin, the founders of Google, to rank web pages in search engine results.
- It measures the importance of a webpage based on the quantity and quality of links pointing to it.

- **Purpose:**

- To determine the relevance and authority of webpages, helping search engines deliver more accurate and valuable search results to users.

# How PageRank Works

- **Basic Concept:**

- Imagine the web as a network of interconnected pages (nodes) linked by hyperlinks (edges).
- Each link from one page to another is considered a "vote" of importance.
- However, not all votes are equal; a vote from a highly ranked page is more valuable than one from a lower-ranked page.

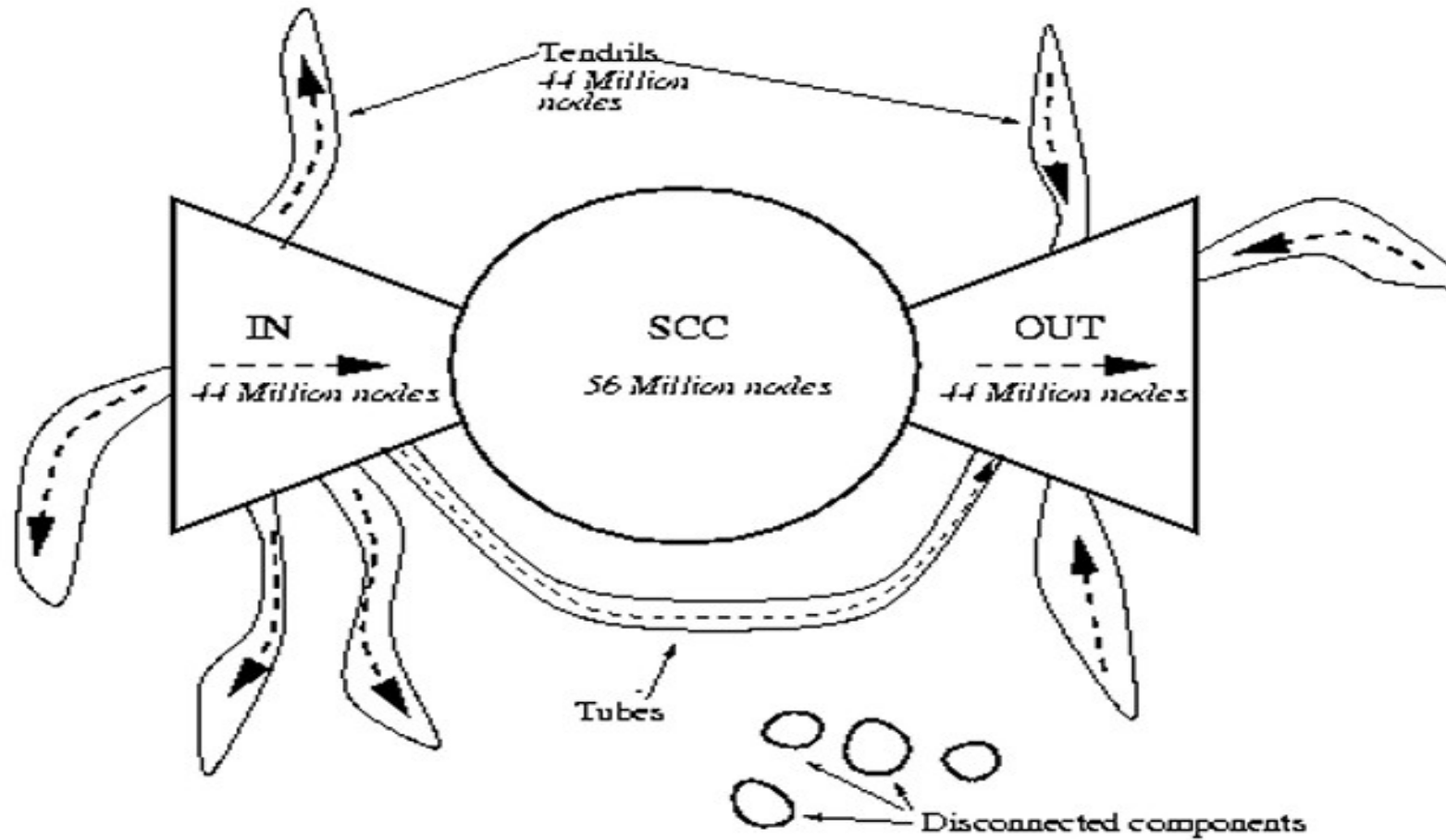
- **Rank Distribution:**

- PageRank distributes a ranking score to each page based on the number and quality of links it receives.
- The algorithm iteratively updates these scores until they stabilize, reflecting the relative importance of each page.

# Example to Illustrate PageRank

- **Scenario:**
  - Consider a small web of 4 pages: **A**, **B**, **C**, and **D**.
- **Link Structure:**
  - **Page A** links to **Page B** and **Page C**.
  - **Page B** links to **Page C**.
  - **Page C** links to **Page A**.
  - **Page D** links to **Page C**.
- **Initial PageRank Assignment:**
  - Each page starts with an equal PageRank score, e.g., 1.0.
- **Iterative Calculation:**
  - **Page A's PageRank:** Receives links from **Page C**.
  - **Page B's PageRank:** Receives links from **Page A**.
  - **Page C's PageRank:** Receives links from **Page A**, **Page B**, and **Page D**.
  - **Page D's PageRank:** Receives no incoming links.
- **First Iteration Results:**
  - **Page A:** New PageRank = Page C's PageRank / Number of outbound links from C.
  - **Page B:** New PageRank = Page A's PageRank / Number of outbound links from A.
  - **Page C:** New PageRank = (Page A's PageRank / outbound links from A) + (Page B's PageRank / outbound links from B) + (Page D's PageRank / outbound links from D).
  - **Page D:** New PageRank = 0 (no incoming links).
- **Convergence:**
  - Repeat the iterative process until PageRank values stabilize, indicating the final ranking.

# Structure of web page



# Dead Ends in the Web

- A **dead end** (or dangling node) occurs when a page has no outbound links. These are pages from which no other page can be reached.
- **Example of a Dead End :**
- Page A links to Page B and Page C.
- Page B links to Page D.
- Page C links to Page A.
- Page D has no outbound links (dead end).

In this case, Page D is a dead end because there is no further path from D to any other page.

## **Solution to Dead Ends:**

Teleportation Factor: To solve the issue of dead ends, PageRank introduces a random surfing factor (often set to 0.85). This allows users to "teleport" to any random page, ensuring PageRank values continue circulating even if dead ends exist.

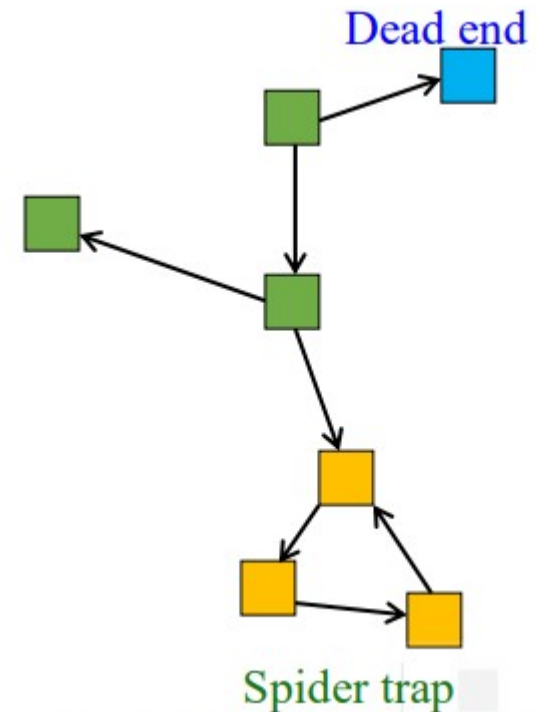


Figure 7: Dead end and Spider trap Problem [2]

# What is a Spider Trap?

- A **Spider Trap** is a subset of web pages that are tightly interlinked among themselves but have few or no links to the rest of the web.
- It creates a closed loop where web crawlers (spiders) can get "trapped" endlessly navigating within the trap without reaching other parts of the web.
- **Characteristics:**
  - **Highly Interconnected:** Pages within the trap link to each other frequently.
  - **Isolation:** Minimal or no outbound links to external pages outside the trap.
  - **Loop Formation:** Forms a loop that can confound crawling algorithms.
- **Example of a Spider Trap**
  - **Page A** links to **Page B** and **Page C**.
  - **Page B** links to **Page A** and **Page D**.
  - **Page C** links to **Page A** and **Page D**.
  - **Page D** links back to **Page B** and **Page C**.
- Pages A, B, C, and D form a closed loop.
- Crawlers moving from A to B to D to C and back to A can cycle indefinitely within this group.

# Solutions to Spider Trap Problem

- **Teleportation Factor (Random Jump):**

- Introduce a probability that the crawler will randomly jump to any page in the web instead of following a link.
- Ensures that rank can flow out of spider traps and distribute across the entire web.

- **Damping Factor:**

- Typically set around 0.85 in PageRank, representing the probability of following a link versus jumping to a random page.
- Helps mitigate the effects of spider traps by allowing rank to escape closed loops.

- **Normalization Techniques:**

- Adjust the PageRank formula to account for the presence of spider traps.
- Ensure that rank is properly distributed even in the presence of tightly interlinked subsets.

# What is Link Spam?

- **Link Spam** refers to the practice of manipulating hyperlinks to artificially inflate the ranking of a webpage in search engine results.
- It involves creating a large number of low-quality or irrelevant links to a target page to deceive search engines like Google.
- **Purpose:**
  - To improve a webpage's visibility and ranking in search engine results pages (SERPs) without providing genuine value or content quality.

## How Link Spam Affects PageRank

- **Impact on PageRank:**
  - PageRank relies on the quantity and quality of inbound links to determine a page's importance.
  - Link spam inflates the number of inbound links, artificially boosting PageRank scores.
- **Consequences:**
  - **Distorted Rankings:** Legitimate pages may be pushed down in search results.
  - **Reduced Search Quality:** Users may encounter less relevant or lower-quality content.
  - **Erosion of Trust:** Search engines may become less reliable if rankings are manipulated.

# Types of Link Spam and Example

- **a. Reciprocal Linking:**
  - **Description:** Websites agree to link to each other to increase their link counts.
  - **Example:** Two blogs mutually linking to each other regardless of content relevance.
- **b. Private Blog Networks (PBNs):**
  - **Description:** A network of websites created solely to link back to a main site to boost its rankings.
  - **Example:** A marketer creates multiple fake websites, each linking to their primary business site.
- **c. Comment Spam:**
  - **Description:** Automated or mass posting of irrelevant comments with backlinks to manipulate SEO.
  - **Example:** Posting the same promotional comment with a link on numerous blog posts or forums.
- **d. Hidden Links:**
  - **Description:** Links are hidden from users but visible to search engines, often using small fonts or matching background colors.
  - **Example:** Text links embedded in a webpage's footer that are invisible to visitors but crawled by search engines.
- **e. Link Farms:**
  - **Description:** Groups of websites that exist solely to provide outbound links to other sites, creating a dense web of spammy links.
  - **Example:** A directory site that lists thousands of unrelated websites purely for link generation.
- **Example of Link Spam**
  - **Website X** wants to rank higher for the keyword "best gardening tools."
  - To achieve this, Website X engages in link spam by:
    - Creating 100 fake websites (PBNs), each linking back to Website X.
    - Posting spammy comments with backlinks to Website X on 500 unrelated blogs.
    - Embedding hidden links to Website X in the footer of numerous low-quality sites.

# What is Topic-Sensitive PageRank?

- **Topic-Sensitive PageRank** is a variant of the traditional PageRank algorithm that personalizes ranking based on specific topics or user interests.
- It assigns different importance scores to pages depending on their relevance to particular topics, enhancing the relevance of search results for diverse queries.
- **Purpose:**
  - To provide more accurate and relevant search results by considering the topical context, thereby improving user satisfaction and search engine performance.

# How Topic-Sensitive PageRank Works

- Traditional PageRank uses a single teleportation (random jump) vector, assuming equal probability of jumping to any page.
- Topic-Sensitive PageRank employs multiple teleportation vectors, each biased towards a specific topic.
- **Implementation Steps:**
  - **Define Topics:** Identify a set of topics relevant to the domain (e.g., sports, technology, health).
  - **Create Seed Sets:** For each topic, select a set of seed pages that are highly relevant to that topic.
  - **Compute Topic Vectors:** Generate teleportation vectors where each vector is biased towards its respective seed set.
  - **Run PageRank:** Perform separate PageRank computations for each topic vector, resulting in multiple PageRank scores per page, each reflecting its importance within a specific topic.

# Example of Topic-Sensitive PageRank

- Imagine a search engine serving queries in two distinct topics: "**Artificial Intelligence**" and "**Healthy Eating**".
- **Step-by-Step Example:**
  - **Define Topics:**
    - **Topic 1:** Artificial Intelligence (AI)
    - **Topic 2:** Healthy Eating
  - **Create Seed Sets:**
    - **AI Seed Pages:** Wikipedia's AI page, leading AI research institutions, prominent AI blogs.
    - **Healthy Eating Seed Pages:** Government health websites, popular nutrition blogs, leading health organizations.
  - **Compute Topic Vectors:**
    - **AI Vector:** High probability of teleporting to AI seed pages.
    - **Healthy Eating Vector:** High probability of teleporting to Healthy Eating seed pages.
  - **Run PageRank:**
    - **AI PageRank:** Pages related to AI receive higher scores within the AI context.
    - **Healthy Eating PageRank:** Pages related to Healthy Eating receive higher scores within the Healthy Eating context.
- **Result:**
  - When a user searches for "machine learning," the AI PageRank scores prioritize relevant AI pages.
  - When a user searches for "balanced diet," the Healthy Eating PageRank scores prioritize relevant health pages

# HUBS AND AUTHORITIES

The basic idea in hubs and authority model is that every web page is assigned two scores, not just one, as is the case of PageRank algorithm.

One type of score is called a hub score and the other one will be called as an authority score.

An authority value is computed as the sum of the scaled hub values that point to a particular web page.

A hub value is the sum of the scaled authority values of the pages it points to.

**The basic features of this algorithm are:**

- The hubs and authority based algorithm is named HITS algorithm, which is Hypertext Induced Topic Selection.
- It is used to measure the importance of pages or documents in a similar way to what we did with PageRank computation.
- For example, we need to find a set of good newspaper webpages.

The idea is that we do not just want to find good newspapers, but in some sense, we want to find specialists (persons) who link in a coordinated way to good newspapers.

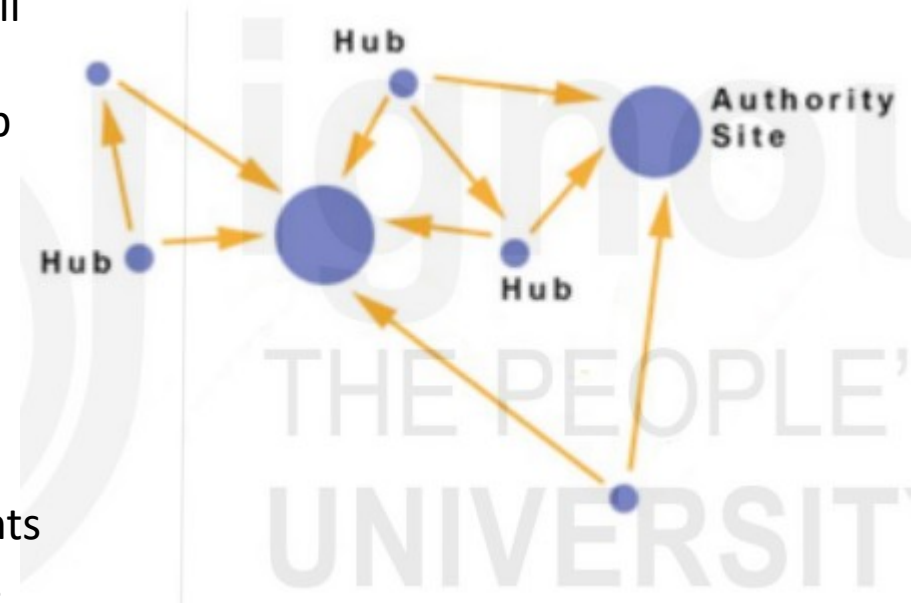


Figure 12: Hubs and Authorities [4]

So, the idea is similar to PageRank such that we will count links as votes.

- For every page, you will have to compute its hub and its authority score.
- Each page will basically have a quality score of its own as an expert. We call this as a hub score
- Each page also has a quality score of it as a content provider, so we call it an authority score.
- So, the idea would be that the hub score is simply the sum of all the votes of the authorities that are pointed to. The authority score will be the sum of the experts votes that the page is receiving.
- We will then apply the principle of repeated improvement to compute the steady state score. Hence, the way we will think about this is that generally pages on the web fall into two classes, hubs and authorities.

**Class of Authorities:** These are the pages which contain useful information or content. So in our case of newspaper example, these are newspaper webpages.

**Class of Hub:** where pages link to good authorities or list of good things on the web. So in newspaper example, these are my favorite newspapers and that page would link to them. The HITS algorithm, thus, minimizes the problem due to link spam and users would be able to get good results, as high-ranking pages

# Social networks as graphs

- In the context of social networks as graphs, **nodes** and **edges** are fundamental components used to model the relationships between entities. The structure of social networks can be naturally represented as a graph, where these elements represent individuals and their connections.

## Fundamental Characteristics of Social Networks as Graphs

### 1. Nodes (Vertices):

1. **Definition:** In a social network graph, **nodes** represent entities such as individuals, organizations, or any other social actors. Each node corresponds to a person or an entity in the social network.
2. **Example:** If the social network represents Facebook, a node could represent a person with an account.

### 2. Edges (Links or Connections):

1. **Definition:** **Edges** represent relationships or interactions between the nodes. They are the links that connect pairs of nodes and signify a connection between them.
2. **Example:** An edge in a Facebook network could represent a "friendship" between two people or a "follower" relationship in a Twitter graph.

### 3. Edges can be:

1. **Directed:** If the relationship is one-way (e.g., follower relationships on Twitter, where one person follows another, but the reverse may not be true).
2. **Undirected:** If the relationship is mutual (e.g., friendships on Facebook, where both parties are considered friends).

# Key Characteristics of Social Networks as Graphs

## 1. Degree:

1. The degree of a node refers to the number of connections it has. In an undirected graph, the degree is the number of edges connected to the node. In a directed graph, nodes can have both in-degree (number of incoming edges) and out-degree (number of outgoing edges).
2. **Example:** In a Facebook friendship graph, if a person (node) has 200 friends, their degree is 200.

## 2. Centrality:

1. Centrality measures the importance or influence of a node within the network. Common types of centrality include:
  1. **Degree Centrality:** Nodes with the most connections.
  2. **Betweenness Centrality:** Nodes that act as bridges between different groups.
  3. **Closeness Centrality:** Nodes that are close to all other nodes in terms of shortest paths.
2. **Example:** A celebrity on Twitter may have high degree centrality due to their large number of followers, while a person who connects two groups of people in the network has high betweenness centrality.

## 3. Clustering:

1. Clustering refers to the tendency of nodes to form tightly-knit groups or communities where members are more connected to each other than to nodes outside the group.
2. **Example:** In a social network of students, a cluster may represent a group of friends who all know each other.

## 4. Path Length:

1. The path length is the number of edges that must be traversed to connect one node to another. The shorter the path length between nodes, the more directly connected they are.
2. **Example:** In LinkedIn, the "degree of separation" between two users reflects the shortest path connecting them through mutual connections.

## 5. Connectedness:

1. A graph is connected if there is a path between any pair of nodes. In a social network, connectedness measures how many people are connected either directly or indirectly to one another.
2. **Example:** A large, tightly connected social network means that most people can reach each other through a few intermediary connections.

## 6. Homophily:

1. Social networks often exhibit homophily, which is the tendency for people to form connections with others who are similar to them in terms of demographics, interests, or behaviors.
2. **Example:** On LinkedIn, professionals in the same industry or role tend to form connections more frequently than those in different fields.

# Real-Life Example of a Social Network as a Graph

- Consider a **Twitter** network. The users are the **nodes** of the graph, and the **edges** represent the "follower" relationships between the users.
- If User A follows User B, there is a **directed edge** from User A (node) to User B (node).
- If User A and User B follow each other, there are **two directed edges**, one from A to B and one from B to A.
- If a group of friends follows each other closely, they would form a **cluster** in the graph.
- A celebrity who has millions of followers would have a high **degree centrality**, as they have many incoming edges (followers).

# Common types of social network graphs:

- Social network graphs can take on various forms depending on the nature of the relationships between the nodes (entities) and the structure of the network. Here are some common types of social network graphs:
- **1. Undirected Graph**
  - In an undirected graph, the edges between nodes have no direction. The relationship is bidirectional, meaning if Node A is connected to Node B, the reverse is also true.
  - **Example:** A Facebook friendship network is an undirected graph because if two people are friends, the relationship is mutual.
- **2. Directed Graph (Digraph)**
  - In a directed graph, the edges have a direction. This means a connection from Node A to Node B does not necessarily imply a connection from Node B to Node A.
  - **Example:** Twitter's follower network is a directed graph, where one person can follow another without being followed back.
- **3. Weighted Graph**
  - In a weighted graph, each edge has a weight associated with it, which represents the strength or significance of the relationship between nodes.
  - **Example:** In a professional network like LinkedIn, the weight on an edge could represent how many interactions (e.g., messages exchanged, endorsements) have occurred between two people.
- **4. Bipartite Graph**
  - A bipartite graph is a special type of graph where nodes can be divided into two distinct sets, and edges only exist between nodes from different sets, not within the same set.
  - **Example:** In a graph representing user-product recommendations, one set of nodes represents users, and the other set represents products. Edges connect users to the products they have purchased or rated.
- **5. Multigraph**
  - A multigraph allows multiple edges between the same set of nodes. These multiple edges can represent different types of relationships between the same entities.
  - **Example:** In a communication network, two people (nodes) might be connected by multiple edges representing different modes of communication (e.g., phone calls, emails, and social media messages).

- **6. Homophily Graph**

- A homophily graph captures the tendency for individuals to connect with others who are similar to them in terms of certain attributes, such as age, profession, or interests.
- **Example:** In a social network where people group based on shared interests, people with similar hobbies or job roles form dense clusters.

- **7. Small-World Network**

- A small-world network is characterized by short paths between any two nodes, meaning most nodes can be reached from others through just a few intermediaries.
- **Example:** Social media networks like Facebook often exhibit small-world properties, where even if two people don't know each other, they can likely be connected through a few mutual friends.

- **8. Scale-Free Network**

- A scale-free network follows a power-law distribution, where a few nodes (called hubs) have a large number of connections, while most nodes have only a few connections.
- **Example:** The internet is a scale-free network where a few websites (like Google, Facebook) receive a large portion of traffic, while many smaller websites have much fewer visitors.

- **9. Dynamic Graph**

- A dynamic graph changes over time as nodes and edges are added or removed. These graphs are often used to model evolving social networks.
- **Example:** In an evolving social network like Instagram, new users (nodes) join the network over time, and connections (edges) between users also change as they follow or unfollow others.

- **10. Clique**

- A clique is a complete subgraph where every node is directly connected to every other node. This type of graph models tightly-knit groups where everyone knows each other.
- **Example:** A close friend group within a larger social network might form a clique where every member has a direct connection to every other member.

## Direct Discovery of Communities

- In the context of **social network analysis**, discovering communities means identifying groups of nodes (people, websites, or entities) that are more closely connected to each other than to the rest of the network.
- **Direct Discovery of Communities** refers to methods that directly identify these groups based on the structure of the network.
- **Real-Life Example: Friend Circles on Social Media**
- Let's use a **social media platform** (like Facebook) to explain **Direct Discovery of Communities**.

### Example: Friend Groups on Facebook

- Imagine a large social network like Facebook. Every user is a node, and every friendship is an edge connecting two nodes. Communities represent **friend circles**—groups of friends who interact more frequently with each other than with others.
1. **Family Group:** You have close ties to your family members, and they often interact with each other too. On the graph, this would look like a cluster where everyone is highly interconnected.
  2. **Work Friends:** At your workplace, you may have a separate circle of connections, with everyone in your company linked together by mutual work relationships.
  3. **College Friends:** Similarly, you have a group of friends from college, and this circle is another community where everyone is closely connected.

### How Communities are Discovered:

4. **Density of Connections:** The algorithm used for direct discovery of communities analyzes the *density of connections* between nodes (people). If a group of people interacts frequently (with lots of mutual connections), they form a community.
5. **Detecting Separation:** The algorithm detects where groups are relatively *disconnected* from the rest of the network. For example, your **college friend circle** may have fewer connections to your **family circle**, so the algorithm can separate these groups into distinct communities.
6. **Clustering Algorithms:** Techniques like **Girvan-Newman** or **Louvain Method** are often used to break down the network into communities by either removing edges or clustering nodes based on connection density.

# Recommendation Systems

**Recommendation Systems** are algorithms designed to suggest relevant items to users based on various data inputs and user interactions.

- **Importance:**
  - Enhance user experience by personalizing content.
  - Drive engagement and sales in platforms like e-commerce, streaming services, and social media.
- **Example:**
- **Netflix:** Recommends movies and TV shows based on your viewing history.
- **Amazon:** Suggests products based on your purchase and browsing behavior.

## Models for Recommendation Systems

- Recommendation systems can be built using various models, each leveraging different types of data and methodologies.
- **Primary Models:**
  - **Content-Based Filtering**
  - **Collaborative Filtering**
  - **Hybrid Models**

# Utility matrix

- Utility matrix is a fundamental concept in recommendation systems used to represent user preferences for items, like movies, products, or services.
- It is a two-dimensional matrix where rows represent users and columns represent items.
- The values in the matrix are either explicit (e.g., user ratings for items) or implicit (e.g., number of purchases or clicks) preferences.
- These values represent how much a user likes or interacts with a particular item.
- **Structure of the Utility Matrix**
- **Rows:** Each row represents a different user.
- **Columns:** Each column represents an item (like a movie, product, etc.).
- **Values:** Each cell contains a numerical value representing the user's interaction or preference for the item.
- A blank or missing value means the user has not interacted with or rated that item.

In this example:

- User 1 has rated Movie A with a score of 4, Movie B with a 5, and Movie D with a 2.
- User 2 hasn't rated Movie A or Movie D, but has given Movie B a score of 3 and Movie C a score of 4.

### Use of Utility Matrix in Recommendation Systems

The utility matrix is used as input for building recommendation models. The goal is to predict the missing values in the matrix and recommend items that a user is likely to enjoy based on their past interactions and the interactions of other users.

#### 1. Content-Based Recommendations:

In content-based recommendation systems, the utility matrix is combined with item metadata (such as genres for movies or product descriptions). The system recommends items that are similar to the ones the user has interacted with or rated highly. For example:

- If User 1 has rated Movie B highly, the system might look at the genre, director, and actors in Movie B and recommend other movies that share similar characteristics

|        | Movie A | Movie B | Movie C | Movie D |
|--------|---------|---------|---------|---------|
| User 1 | 4       | 5       |         | 2       |
| User 2 |         | 3       | 4       |         |
| User 3 | 5       |         | 2       | 1       |
| User 4 | 3       | 2       | 5       |         |

- **2. Collaborative Filtering:**

- Collaborative filtering focuses on the utility matrix itself, using either user-based or item-based filtering methods to make predictions.
- **User-based collaborative filtering:** The system finds users who have similar taste to the target user (based on overlapping preferences) and recommends items that those similar users liked.
- Example:
  - If User 1 and User 4 have both rated Movie A and Movie B similarly, and User 4 has also rated Movie C highly, the system might recommend Movie C to User 1.
- **Item-based collaborative filtering:** The system identifies items that are similar based on user ratings and recommends items similar to those the target user has liked.
- Example:
  - If Movie A and Movie B have been rated similarly by multiple users, the system might recommend Movie B to a user who liked Movie A.

## **Challenges in Utility Matrix**

- 1. Sparsity:** Utility matrices are often sparse because most users have not interacted with most items. This makes it challenging to make accurate recommendations.
- 2. Cold Start Problem:** New users or new items lack enough data in the utility matrix, making it difficult to provide recommendations for them.
- 3. Scalability:** As the number of users and items grows, the size of the utility matrix increases, making it computationally expensive to process

# Content-Based Recommendations

**Content-Based Filtering** recommends items similar to those a user has liked in the past, based on item features.

- **How It Works:**
  - **Item Profiles:** Each item is described by a set of features (e.g., genre, keywords).
  - **User Profiles:** Created based on the features of items the user has interacted with.
  - **Similarity Calculation:** Items are recommended based on the similarity between item profiles and user profiles.
- **Advantages:**
  - **Personalization:** Tailors recommendations to individual preferences.
  - **No Cold Start for Items:** New items can be recommended if their features are known.
- **Limitations:**
  - **Limited Discovery:** May not introduce diverse or novel items.
  - **Feature Engineering:** Requires detailed and accurate item descriptions.

## Example:

- **Spotify:** Recommends songs with similar genres, tempos, and instrumentation to those you've previously listened to.

# Collaborative Filtering

**Collaborative Filtering** recommends items based on the preferences of similar users, without requiring item feature information.

- **Types:**
  - **User-Based Collaborative Filtering**
    - Finds users similar to the target user and recommends items they liked.
  - **Item-Based Collaborative Filtering**
    - Finds items similar to those the target user has liked and recommends those.
- **How It Works:**
  - **User Similarity:** Calculated using metrics like cosine similarity or Pearson correlation.
  - **Prediction:** Aggregates the preferences of similar users to predict the target user's preference for items.
- **Advantages:**
  - **No Need for Item Features:** Works purely on user interaction data.
  - **Discover Novel Items:** Can recommend items outside the user's usual preferences.
- **Limitations:**
  - **Cold Start for Users:** New users with no history are difficult to recommend for.
  - **Scalability:** Can be computationally intensive with large datasets.
- **Example:**
  - **Amazon:** Recommends products that customers with similar purchase histories have bought.

# Hybrid Recommendation Systems

**Hybrid Systems** combine multiple recommendation techniques (e.g., content-based and collaborative filtering) to leverage their strengths and mitigate their weaknesses.

- **Advantages:**

- **Improved Accuracy:** Combines diverse data sources for better predictions.
- **Enhanced Coverage:** Addresses cold start and sparsity issues.

- **Methods:**

- **Weighted Hybrid:** Assigns different weights to different recommendation sources.
- **Switching Hybrid:** Switches between different methods based on context.
- **Mixed Hybrid:** Merges recommendations from multiple methods into a single list.
- **Feature Combination:** Integrates features from different models into a unified framework.

- **Example:**

- **Netflix:** Uses both collaborative filtering (user viewing patterns) and content-based methods (movie genres, descriptions) to recommend shows.

### **Challenges:**

- **Data Privacy:** Ensuring user data is protected while providing personalized recommendations.
- **Scalability:** Handling large-scale data efficiently.
- **Bias and Fairness:** Avoiding algorithmic biases that can skew recommendations.

### **Future Directions:**

- **Deep Learning:** Leveraging neural networks for more accurate recommendations.
- **Context-Aware Recommendations:** Incorporating contextual information like time, location, and device.
- **Explainable AI:** Providing transparent explanations for recommendations to build user trust.

# Model for Recommendation Systems

Recommendation systems are algorithms designed to suggest relevant items to users based on various data inputs, enhancing user experience by personalizing content delivery

## Components of Recommendation Systems:

### 1. Data Collection:

1. **User Data:** User profiles, preferences, and behavior (e.g., purchase history, viewing habits).
2. **Item Data:** Attributes and metadata of items (e.g., genre, price, description).
3. **Interaction Data:** Explicit feedback (ratings, likes) and implicit feedback (clicks, time spent).

### 2. Data Processing:

1. **Preprocessing:** Cleaning and normalizing data.
2. **Feature Extraction:** Identifying key attributes from item and user data.

### 3. Modeling:

1. **Algorithm Selection:** Choosing appropriate recommendation algorithms (content-based, collaborative filtering, hybrid).
2. **Training:** Using historical data to train the model.

### 4. Recommendation Generation:

1. **Scoring:** Calculating relevance scores for items.
2. **Ranking:** Ordering items based on scores to present top recommendations.

### 5. Evaluation:

1. **Metrics:** Precision, recall, F1-score, RMSE (Root Mean Square Error).
2. **A/B Testing:** Comparing different recommendation strategies.

### • Example:

### • Amazon's Recommendation Engine:

- **Data Collection:** Tracks user's browsing history, purchase history, and ratings.
- **Modeling:** Uses collaborative filtering to find users with similar purchasing patterns.
- **Recommendation Generation:** Suggests products that similar users have bought but the current user hasn't.

# Content-Based Recommendations

Content-Based Filtering recommends items similar to those a user has liked in the past by analyzing the attributes and features of items and matching them with the user's preferences.

## 1. Item Profiling:

1. Each item is described using a set of features. For movies, features might include genre, director, actors, keywords, etc.

## 2. User Profiling:

1. A user profile is built based on the features of the items they have interacted with (liked, rated highly, purchased).

## 3. Similarity Measurement:

1. The system calculates the similarity between the user profile and potential items using metrics like cosine similarity, Euclidean distance, or Jaccard similarity.

## 4. Recommendation Generation:

1. Items with the highest similarity scores to the user profile are recommended.

### • Techniques:

- **Vector Space Model:** Represents items and user profiles as vectors in a multi-dimensional space.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Weighs the importance of features based on their frequency across items.

### Advantages:

- **Personalization:** Tailors recommendations to individual user preferences.
- **Transparency:** Easy to understand why an item was recommended based on its features.
- **No Cold Start for Items:** New items can be recommended as long as their features are known.

### Limitations:

- **Limited Novelty:** May only recommend items similar to what the user already knows, limiting discovery of new interests.
- **Feature Dependence:** Requires comprehensive and accurate feature descriptions of items.
- **Over-Specialization:** Recommendations can become too narrow, ignoring broader user interests.

### Example:

#### • Spotify's Music Recommendations:

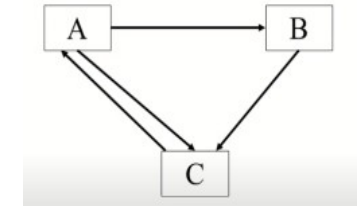
- **Item Features:** Genre, tempo, key, instrumentation, and mood.
- **User Profile:** Based on the features of songs the user frequently listens to.
- **Recommendation:** Suggests new songs with similar attributes to those the user like

# Important concepts

- dead ends in the context of link analysis on the web
- difference between content-based recommendations and collaborative filtering in recommendation engines
- clustering in the context of social network graphs
- Common algorithm used for clustering social network graphs.
- dangling node in a web graph
- common types of social network graphs
- utility matrix with example
- Explain how PageRank is used in search engines to rank web pages. What does a higher PageRank indicate?

- concept of the PageRank algorithm and evaluate its significance in web search and ranking of web pages.
- structure of web and spider trap
- fundamental characteristics that define social networks as graphs. What are nodes and edges in this context
- How does the process of direct discovery of communities work in the analysis of social network graphs?
- primary objective of recommendation systems, and why are they valuable for businesses and users
- Compare and contrast the direct discovery of communities in social network graphs with other community detection algorithms. What are the strengths and weaknesses of each approach
- Concept of topic-sensitive PageRank and how it differs from the standard PageRank algorithm.

- Compute the page rank of each page in the following graphs after 3 iterations



- principles of link spam detection. strategies used by link spammers, and how can PageRank be used to identify them
- Apply different types of recommendation systems (e.g., content-based, collaborative filtering, hybrid) to recommend products in an online bookstore. Provide specific examples and discuss the expected benefits and limitations of each type in this context.
- the importance of hubs and authorities in link analysis. How do these concepts relate to the HITS (Hyperlink-Induced Topic Search) algorithm
- Compute hub and authority score for the matrix using the HITS algorithm with  $k=3$  and identify the best authority and hub node

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 |

- Consider the Web graph given below with six pages (A, B, C, D, E, F) with directed links as follows:
  - $A \rightarrow B, C$
  - $B \rightarrow A, D, E, F$
  - $C \rightarrow A, F$
 Assume that the PageRank values for any page  $m$  at iteration 0 is  $PR(m) = 1$  and teleportation factor for iterations is  $\beta = 0.85$ . Perform the PageRank algorithm and determine the rank for every page at iteration 2.
- Apply the principles of collaborative filtering to recommend movies to a user based on their previous movie ratings and the ratings of other similar users.